

1 Preliminary Title

Working Title: **Asymmetric and Nonlinear Effects of Business-Cycle Shocks on Bank Size: Lingering Impacts of Macroeconomic Shocks on Bank Growth.**

2 Research Question and Motivation

Research Question:

How do business-cycle shocks impact bank size, and does the direction of these shocks have different outcomes for varying bank categories?

This question is motivated by the literature on income and wealth inequality, particularly research examining the distributional consequences of macroeconomic shocks in the United States. A substantial body of evidence demonstrates that individual wealth is highly concentrated among a relatively small share of the population. A related, but comparatively less discussed, form of concentration exists within the U.S. banking sector: a disproportionate share of total banking assets is held by a relatively small number of institutions.

As of the first quarter of 2026, 4,289 U.S. banks reported to the Federal Deposit Insurance Corporation (FDIC). The distribution of bank assets is highly right-skewed, as illustrated by the Lorenz curve in Figure 1, indicating substantial concentration in the sector. Research on income and wealth inequality suggests that macroeconomic shocks do not affect all segments of the distribution uniformly; rather, their effects may vary across wealth quantiles. This study extends that perspective to the banking sector by examining whether banks of different sizes respond differently to business-cycle fluctuations. Specifically, it investigates whether changes in quarterly real GDP have heterogeneous, nonlinear, or asymmetric effects on bank size, and whether negative shocks generate more persistent consequences than positive shocks.

The paper's central hypothesis is that macroeconomic shocks affect banks in nonlinear and asymmetric ways similar to those documented for individuals [1]. Specifically, we hypothesize that negative shocks generate larger and more persistent reductions in bank growth than the increases generated by positive shocks. Furthermore, analogous to the greater sensitivity of individuals at the upper end of the earnings distribution to economic downturns, we expect the largest banks to exhibit more negatively skewed response distributions than their smaller counterparts. We expect the negative effects of macroeconomic shocks on banks to be less pronounced than those experienced by individuals. This reflects the fact that severe bank contractions or failures are considerably less common than individual job loss, and banks may possess greater access to financial buffers, federal government support, and risk-management tools that mitigate the effects of adverse shocks.

The mechanisms creating these bank-dependent heterogeneous effects differ from those impacting individuals. Adverse macroeconomic conditions may increase defaults and lower loan demand, weakening banks over several quarters. Banks may also become more risk-averse during downturns, choosing not to lend to higher-risk individuals. The largest banks may be particularly affected by recessions as their lending composition includes massive loans and lines of credit to the largest business, and a single default can shock the balance sheet. Alternative mechanisms include bank survivorship bias as the cause of the asymmetric effects, larger banks diversification and hedging

tools as a method to resist recessions, and Federal Reserve policy as the main driver of loan growth or decay.

3 Literature Review

A substantial literature examines the effects of macroeconomic shocks on individual earnings distributions and bank health; however, relatively little research addresses the specific relationship investigated in this paper. The analytical framework developed for earnings dynamics can be adapted to study banks, making the proposed analysis both novel and empirically feasible. Furthermore, the project provides an opportunity to apply recent advances in heterogeneous-agent modeling to the analysis of business-cycle dynamics [2]. The paper will be broken into three main sections, each relevant to a particular vein of literature.

First, **Güvenen et al. (2021)** and **Güvenen, Ozkan, and Song (2014)** establish the mechanisms underlying changes in household income over the business cycle. This paper adapts their framework to the banking sector by examining how bank growth responds to macroeconomic conditions. Whereas age represents an important source of heterogeneity in household income and wealth dynamics, bank-specific characteristics—including size, leverage, efficiency, and capital holdings—will be evaluated to determine which factors most strongly explain differences in banks' responses to macroeconomic shocks [3] [1].

Next, **Güvenen, McKay, and Ryan (2026)** and **Francis (2026)** provide examples of potential modeling frameworks. Traditionally, heterogeneous-agent models of households are based on the Bewley–Huggett–Aiyagari framework, also known as the standard incomplete-markets model. These models are typically calibrated using household survey data to reproduce key second-moment properties of the income and wealth distributions. This paper adapts that framework to the banking sector by treating banks as the heterogeneous agents and incorporating bank-specific constraints into their optimization problem. The survey data and specific constraints will be shown in the next sections [2] [4].

Finally, **Corbae and D'Erasmus (2021)** provides a heterogeneous-agent model of the banking sector. Their framework examines how regulatory policies influence banks' risk-taking behavior and market structure, whereas this paper focuses on how macroeconomic conditions shape banks' growth, risk-taking, and distributional outcomes. Thus, the proposed analysis extends this literature by treating macroeconomic fluctuations as the primary driver of heterogeneous bank responses [5].

4 Data Description

Estimating the proposed heterogeneous-bank model requires a substantial amount of data, particularly because the analysis relies on accurate estimates of the third and fourth moments of bank-level outcomes.

Fortunately, the FDIC BankFind Suite provides quarterly financial records for FDIC-insured institutions beginning in the first quarter of 1992. This paper will use data from 2005 through 2025, a period that encompasses multiple banking crises, recessions, and expansions. Because the data are provided separately for each quarter, the files must be consolidated into a single panel dataset. This

data-construction process has already been completed, and the resulting dataset includes quarterly observations on bank size, efficiency ratios, net interest margins, net interest income, liabilities, and other publicly reported financial measures. With several thousand banks represented in many quarters, the combined panel contains a sufficiently large number of observations to estimate the relevant third- and fourth-moment properties of the distribution. The panel structure controls for bank and quarter fixed effects, thereby accounting for unobserved heterogeneity across banks and common time-specific shocks.

Macroeconomic shocks will be quantified by the quarterly change in real GDP provided by Federal Reserve Economic Data. This is also reported quarterly and is easy to match onto the compiled FDIC data. Large downturns in real GDP are indicative of recessionary behavior, with raises being representative of growth.

5 Proposed Methodology

5.1 Panel Regression Modeling

The basic empirical model from which our heterogeneous bank model will be compared to is a linear, symmetric process:

$$\Delta \log(A_{i,t}) = \alpha_i + \rho \Delta \log(A_{i,t-1}) + \beta \Delta \log(Y_t) + \varepsilon_{i,t}, \quad (1)$$

Here, $A_{i,t}$ denotes the total assets of bank i in period t , while Y_t denotes real GDP in period t . The term $\Delta \log(A_{i,t})$ represents the growth rate of bank assets, and $\Delta \log(Y_t)$ represents real GDP growth. The parameter ρ captures persistence in bank-asset growth, while β measures the response of bank-asset growth to changes in real GDP. The term α_i denotes a bank-specific fixed effect, and $\varepsilon_{i,t}$ represents an i.i.d. normal error term.

An alternative to the model proposed in Equation 1 is a panel regression that captures the impact of real GDP growth and decline separately:

$$g_{i,t} = \alpha_i + \rho g_{i,t-1} + \beta_+ g_t^+ - \beta_- |g_t^-| - \gamma (|g_t^-| \times \log A_{i,t-1}) + \varepsilon_{i,t}. \quad (2)$$

$$g_{i,t} = \Delta \log(A_{i,t}), \quad (3)$$

$$g_t^+ = \max\{\Delta \log(Y_t), 0\}, \quad (4)$$

$$g_t^- = \min\{\Delta \log(Y_t), 0\}. \quad (5)$$

Equation 2 provides a simple method for isolating the impact of real GDP growth versus declines and, particularly, if recessionary periods more negatively impact larger banks. If the hypothesis that economic declines more negatively affect bank size, then $\beta_- > \beta_+$. And, if it is true that these declines are more challenging for large banks, then $\gamma > 0$. What remains to be shown is a rigorous model to evaluate how effects linger past one quarter and which bank-specific variables play the largest role in protecting a banks total asset number during recessionary periods.

The largest direct benefit of the models laid out in Equations 1 and 2 is the ease of computing and back-testing. One CSV file containing FDIC BankSuite Data matched to quarterly real GDP can be imported to STATA and run smoothly. Separating data into training, validation, and testing sets

allows for clear back-testing and usage of a Diebold-Mariano test to determine the more accurate model.

5.2 Basic HANK Framework

The main interest of this paper is to create a heterogeneous banking model that takes into account various properties that makes each individual bank unique. For instance, drawing from **Corbae and D’Erasmus (2021)**, the end goal would be to determine a value function, apply constraints, and solve. Consider the simple Bellman equation below:

$$V_i(A_i, K_i, Z, g^Y) = \max_{A'_i} \left\{ \Pi_i + \beta \mathbb{E} \left[V_i(A'_i, K'_i, Z', g^{Y'}) \right] \right\}, \quad (6)$$

The model consists of heterogeneous banks indexed by $i \in \mathcal{I}$. Each bank is characterized by a vector of state variables, including assets A_i , capital K_i , an idiosyncratic productivity shock Z , and aggregate economic conditions summarized by GDP growth, g^Y . Π_i denotes current-period profits, $\beta \in (0, 1)$ is the discount factor, and expectations are taken over future realizations of both idiosyncratic and aggregate shocks.

This creates a heterogeneous bank model due to the different state variables for each banks. Bank with different asset holdings, capital, or productivity will respond to real GDP shocks differently. The crux of this paper will be centered on determining a realistic value function and constraints to give a theoretical model that can be compared to the regression methods of the previous section.

6 Figures

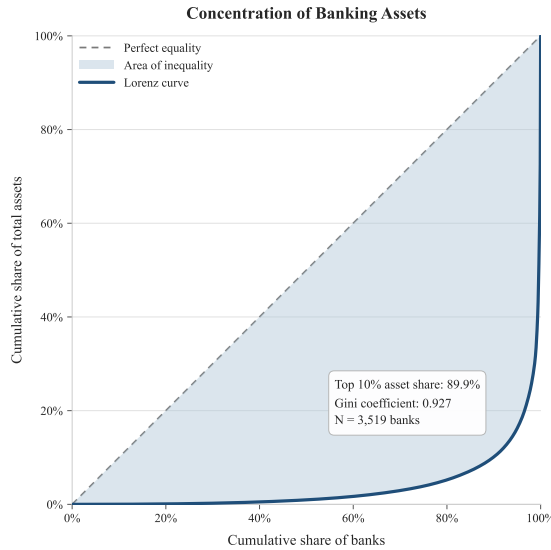


Figure 1: Concentration of banking assets. The Lorenz curve is calculated after excluding banks with assets below \$10,000. The dashed line represents perfect equality, while the shaded region represents the area of inequality.

Relevant Papers

- [1] Fatih Guvenen, Serdar Ozkan, and Jae Song. “The Nature of Countercyclical Income Risk”. en. In: *Journal of Political Economy* 122.3 (June 2014), pp. 621–660. ISSN: 0022-3808, 1537-534X. DOI: 10.1086/675535. URL: <https://www.journals.uchicago.edu/doi/10.1086/675535> (visited on 08/27/2026).
- [2] Fatih Guvenen, Alisdair McKay, and Conor Ryan. “A tractable income process for business cycle analysis”. In: *Review of Economic Dynamics* 61 (Aug. 2026), p. 101345. ISSN: 1094-2025. DOI: 10.1016/j.red.2026.101345. URL: <https://www.sciencedirect.com/science/article/pii/S1094202526000244> (visited on 09/06/2026).
- [3] Fatih Guvenen et al. “What Do Data on Millions of U.S. Workers Reveal About Lifecycle Earnings Dynamics?” en. In: *Econometrica* 89.5 (2021), pp. 2303–2339. ISSN: 1468-0262. DOI: 10.3982/ECTA14603. URL: <https://onlinelibrary.wiley.com/doi/abs/10.3982/ECTA14603> (visited on 09/01/2026).
- [4] Neville Francis. “The Perceived Inflation Wedge.” In: *NBER Working Paper 35354* (2026).
- [5] Dean Corbae and Pablo D’Erasmo. “Capital Buffers in a Quantitative Model of Banking Industry Dynamics”. en. In: *Econometrica* 89.6 (2021), pp. 2975–3023. ISSN: 1468-0262. DOI: 10.3982/ECTA16930. URL: <https://onlinelibrary.wiley.com/doi/abs/10.3982/ECTA16930> (visited on 09/06/2026).

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